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NSP/004/103 (OHI 3) Guidance on the type and fabrication of O/H line supports

1. Purpose

The purpose of this document is to provide guidance on the type and fabrication of overhead line supports for use on the CE Electric UK distribution system.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
OHI 3	April 1998	OHI 3 Pole Types, Specification, Classification and dressing.
RTN / 001 / 500 / 701 / 007	Feb 2000	GN 70-07 Guidance Note for Overhead Line Supports

2. Scope

This document provides guidance on available pole types, including their fabrication, dimensions and dressing criteria.

3. Technical Specification

3.1 Supports types and preservatives

Overhead line supports shall normally be constructed from wood poles manufactured and supplied in accordance with ENA TS 43-88 and NPS/001/001.

- All wood poles for use on the HV distribution system shall be treated with creosote preservative.
- All wood poles for use on the LV distribution system shall be treated with a waterborne preservative Copper Chromic Arsenic) CCA. (CCA gives the poles a greenish appearance)
- Tubular steel poles manufactured and supplied in accordance with NPS/001/018 may be utilised as an alternative type of support on LV unstayed angle/terminal situations.

New wood pole grades and heights will be restricted to the following range:

LV CCA treated poles

9.0 – 14.0m poles in Medium & Stout grade (full height increments only)

HV Creosoted treated poles

9.0 – 14.0m poles in medium grade (full height increments only)

Note.

Where taller medium grade poles are required, this requirement shall be satisfied by the use of an equivalent Stout class pole.

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9.0 – 18.0m poles in stout grade (full height increments only)
9.0 – 18.0m poles in extra stout grade (full height increments only)

LV Steel Poles

9.75m – K class pole

3.2 Fabrication of Poles

All poles shall be drilled, scarfed and notched by the pole supplier in accordance with the company's pole fabrication drawings. New poles shall not be fabricated on site.

Standard fabrication drawings associated with Intermediate structures

Construction	Pole fabrication drawings associated with standard line poles.
LV ABC & Open Wire Construction	1091193301/3
HV – 11/20 & 33kV Construction	1000434001 & 1000434002
CE/C/37 – 66kV Portal Construction	Y204 L0601 Y204 L0602
OHL 4 – 66kV Single Pole Construction	1091231186 1000434002
OHL 9 – 66/132kV Single pole Construction	1091380006 sht2 1091380003 sht2

Where it is required to erect pole mounted plant, the approved pole arrangements shall be selected from a range of pre-fabricated drawing arrangements specified in NSP/004/120.

Poles shall only be fabricated on site where it is required to install equipment on existing poles.

- When scarfing a pole on site, as little timber as possible shall be removed, i.e. only sufficient to meet the requirements of the pole drawing.
- The bottom end of the scarfing must be chamfered to prevent formation of water trap.
- When notching a pole on site the dimensions of the notch are governed by the appropriate pole fabrication drawing or general arrangement drawing.
- Care must be taken, especially with back to back steelwork that the pole is notched to equal depth on each side.
- Pole drillings shall allow for 2mm clearance for all fixing bolts.
- All unused holes shall be plugged with creosote impregnated plugs – see drawing 1091010247 for details.

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3.3 Standard Pole dimensions

Poles dimensions specified within ENA TS 43-88 are designed to be in accordance with BS 1990. The following table details the typical pole dimensions and their associated strengths.

Note

To establish the maximum available strut capability, the figures given in the table must be divided by the appropriate factors of safety as stated in the overhead line design specifications for the particular overhead line specification.

Typical wood pole sizes and strengths – BS1990

Length /	Grade	Min. Top	Max. Top	Dia. 1.5m	Planting	Ult. load	'Single
(m)		Dia. (mm)	Dia. (mm)	Butt (mm)	Depth (mm)	0.6m from Pole Top (kN)	Pole Strut Strength (kN)
9	M	150	180	220	1900	8.15	73.18
10	M	150	185	230	1900	8.13	60.65
11	M	150	190	240	1900	8.18	51.76
12	M	150	190	250	2100	8.23	46.41
13	M	160	200	260	2100	8.46	47.21
14	M	160	205	275	2100	8.98	43.89
9	S	190	240	275	1900	15.91	184.00
10	S	190	245	285	1900	15.47	150.04
11	S	190	250	295	1900	15.24	126.09
12	S	190	250	305	2100	15.08	111.47
13	S	195	255	320	2100	15.73	106.06
14	S	195	255	335	2100	16.24	96.75
15	S	195	260	350	2400	16.68	91.95
16	S	200	265	365	2400	17.35	90.14
17	S	200	265	375	2400	17.2	82.40
18	S	200	265	390	2400	17.57	77.92
20	S	200	265	415	3000	17.74	71.81
22	S	200	265	435	3000	17.49	63.43
24	S	200	265	470	3000	18.36	61.17
11	E/S	245	270	325	1900	TBA	264.78
12	E/S	245	270	345	2100	TBA	244.78
13	E/S	245	270	345	2100	TBA	201.69
14	E/S	245	275	360	2100	TBA	181.79
15	E/S	245	275	375	2400	TBA	171.67
16	E/S	250	280	390	2400	TBA	165.13
17	E/S	250	280	400	2400	TBA	149.83
18	E/S	250	280	415	2400	TBA	140.20

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3.4 Dressing Poles

3.4.1 Crossarm Assemblies:

The main function of the crossarm is to provide sufficient separation of the conductors to allow as long a span length as is practicable without conductor clashing. It also has to have adequate strength to support the loads transmitted to it by either the tension or the weight of the conductor.

Crossarm assemblies are manufactured from mild steel channels or angles which are hot dip galvanised after all cutting and drilling operations have been completed.

- Bolted assemblies are used on ESI 43-40 constructions.
- CE/C/37 (M) crossarms are a welded and bolted assembly.
- The crossarm of CE/OM/DC 1(M) construction has a bolted assembly, but these crossarms also have a wooden insulator insert which provides electrical separation of the two circuits.
- OHL 4 construction crossarms are bolted assemblies

Intermediate crossarms are designed to support conductor weight (including the apparent weight caused by conductor down pull and ice loading).

Section and angle crossarms also have to withstand the forces transmitted to them by conductor tension.

They are normally longer than intermediate crossarms so that adequate conductor spacing is maintained even at the most severe angle positions. To accommodate these two requirements they are manufactured from heavier gauge steel.

Terminal crossarms are similar in length to intermediate crossarms, but since the horizontal loads caused by conductor tension are greatest at these positions the heaviest gauges of steel angles are necessary.

To ensure maximum service life it is essential that the protective galvanising layer is not damaged during site assembly.

3.4.2 Nuts, Bolts & Washers

Metric nuts, bolts and washers are used to assemble all support structures. All components are hot dip galvanised during manufacture.

- The length of a bolt should be such that, after installation, there is a minimum of two full threads projecting from the nut. In situations where the length of bolt is trimmed then the cut end should be coated with a bituminous paint.
- All bolts are fitted with one full nut. Where bolts support a conductor in tension an additional nut is fitted as a locknut.
- Washers are not required for circular holes on metal surfaces, except in the flange of channels when tapered washers are required, or when earthing connections are to be made.
 - o Round flat washers are required for slotted holes on metal surfaces.
 - o Square flat washers are required for flat wooden surfaces.
 - o Square curved washers are required for curved wooden surfaces.

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3.5 Disposal of CCA Poles

Tanalith is used to protect timber from long term fungal and insect attack. It is a waterborne wood preservative which contains copper, chromium and arsenic (CCA). Special precautions need to be taken when handling the chemical during treatment but after impregnation into the timber the chemical becomes fixed into the wood structure and cannot be easily removed.

There are no documented risks associated with drilling and cutting tanalised timber. The sawdust and swarf are harmless and no special precautions are necessary.

Fire Damaged Timber

The only way of releasing the CCA back into its original salt form is by burning. The ash of CCA treated timber contains soluble CCA salts which are hazardous. Therefore when a pole has been damaged by fire the following precautions must be taken.

When handling carbonised timber, protective gloves and overalls must be worn and thorough personal hygiene must be taken.

If the pole is to be removed then to minimise possible dispersal of ash, the affected sections of the pole must be wrapped with polythene sheeting suitably secured in place. If the damage is only slight such that the pole can be left in service then taking the precautions described above, as much carbonised material as possible should be removed and sealed in plastic bags and properly disposed of at licensed waste disposal sites.

Disposal

Tanalised timber must not be used as fuel for barbecues, cooking stoves or grates. Neither must it be given to farmers. It shall be disposed of by cutting it into smaller lengths and sent to registered waste disposal sites.

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4. References

4.1 External Documentation

Reference	Title
BS 1990-1:1984	Wood Poles for overhead power and telecommunications lines
ENA TS 43-88 – part 1	Selection and treatment of Wood Poles and associated Timber for Overhead Lines.

4.2 Internal documentation

Reference	Title
NPS/001/001	Technical specification for wood poles and associated products for overhead lines
NPS/001/018	Technical specification for steel poles for LV overhead lines
NSP/004/120	Guidance notes on the erection of pole mounted plant and equipment.

5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager	G Hammel	31/05/06

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	05/06/06

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	05/06/06

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	21/07/06

6.5 Approval

Approval is given for the content of this document

		Sign	Date
D Sillito	Major Project Manager South	D Sillito	17/06/06
S Gardner	OHL Construction Manager	S Gardner	16/06/06

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
G Earl	Primary Engineering Projects Manager	G Earl	19/06/06